



Q How does my baby's skeleton develop while she's in the uterus?

What begins as a simple framework of cartilage to protect your baby's body hardens into bone through a gradual process of ossification that continues even after your baby is born.

Developing a supportive skeleton

Bones and muscles form early in pregnancy. At five weeks pregnant, your embryonic baby develops a sort of skeletal meshwork of connective tissue (collagen) that protects her rapidly developing organs. Over the next few weeks, this meshwork gets a blood supply, and bone cells called osteoblasts work to toughen it and form cartilage. At 10 weeks, your baby has a recognizable jaw, collarbone, shoulder blades, ribs, vertebrae, and arm and leg bones. Gradually, the osteoblast cells deposit calcium salts on the cartilage to harden, shape, and grow the bones. This process is called ossification and it continues throughout your baby's time in the uterus, and beyond until adulthood when her bones finally stop growing.

Your baby develops muscle tissue early, too: by 10 weeks she can make simple movements. By 20 to 24 weeks her nervous system is becoming connected and her musculoskeletal system has developed enough for her to bend and flex her limbs. In the third trimester of pregnancy, your baby uses 250–350mg of calcium per day to continue developing her bones. Muscle mass increases rapidly, and your baby might double in weight in the last 10 weeks of your pregnancy.

Body structure At 14 weeks, the fetal bones have begun to “ossify” from cartilage to bone, giving your baby enough strength to hold up and move her limbs.

